

UT	ARIES		VENUS -4.8		MARS +1.2		JUPITER -1.9		SATURN +0.4		STARS				
	GHA		GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Name	SHA	Dec		
00	2 50.7		150 28.3	S 19 47.6	243 01.3	N21 31.6	221 55.1	N15 53.6	350 38.5	N 2 18.0	Acamar	315 10.4	S 40 11.5		
01	17 53.2		165 30.0	48.2	258 02.2	31.4	236 57.1	53.4	5 41.2	17.9	Achernar	335 18.6	S 57 05.8		
02	32 55.6		180 31.8	48.7	273 03.1	31.1	251 59.1	53.3	20 43.8	17.8	Acrux	172 59.2	S 63 14.9		
03	47 58.1		195 33.5	49.3	288 03.9	30.9	267 01.1	53.2	35 46.4	17.7	Adhara	255 04.8	S 29 00.2		
04	63 00.6		210 35.3	49.8	303 04.8	30.6	282 03.1	53.0	50 49.1	17.7	Aldebaran	290 37.9	N16 33.8		
05	78 03.0		225 37.1	50.3	318 05.7	30.4	297 05.1	52.9	65 51.7	17.6					
06	93 05.5		240 38.8	S 19 50.9	333 06.6	N21 30.1	312 07.1	N15 52.7	80 54.4	N 2 17.5	Alioth	166 12.3	N55 48.9		
07	108 08.0		255 40.6	51.4	348 07.5	29.8	327 09.1	52.6	95 57.0	17.4	Alkaid	152 51.3	N49 10.9		
08	123 10.4		270 42.4	52.0	3 08.4	29.6	342 11.1	52.5	110 59.6	17.4	Al Na'ir	27 30.7	S 46 49.9		
09	138 12.9		285 44.2	52.5	18 09.3	29.3	357 13.1	52.3	126 02.3	17.3	Alnilam	275 36.3	S 1 10.9		
10	153 15.4		300 46.0	53.0	33 10.1	29.1	12 15.1	52.2	141 04.9	17.2	Alphard	217 46.6	S 8 46.3		
11	168 17.8		315 47.8	53.6	48 11.0	28.8	27 17.1	52.0	156 07.6	17.1					
12	183 20.3		330 49.5	S 19 54.1	63 11.9	N21 28.6	42 19.1	N15 51.9	171 10.2	N 2 17.0	Alphecca	126 02.8	N26 37.7		
13	198 22.8		345 51.3	54.6	78 12.8	28.3	57 21.1	51.8	186 12.8	17.0	Alpheratz	357 33.0	N29 14.5		
14	213 25.2		0 53.1	55.2	93 13.7	28.0	72 23.1	51.6	201 15.5	16.9	Altair	61 58.4	N 8 56.5		
15	228 27.7		15 54.9	55.7	108 14.6	27.8	87 25.1	51.5	216 18.1	16.8	Ankaa	353 05.3	S 42 09.5		
16	243 30.1		30 56.7	56.2	123 15.5	27.5	102 27.1	51.3	231 20.8	16.7	Antares	112 14.2	S 26 29.5		
17	258 32.6		45 58.6	56.7	138 16.4	27.3	117 29.1	51.2	246 23.4	16.7					
18	273 35.1		61 00.4	S 19 57.3	153 17.3	N21 27.0	132 31.1	N15 51.1	261 26.0	N 2 16.6	Arcturus	145 46.9	N19 02.7		
19	288 37.5		76 02.2	57.8	168 18.2	26.8	147 33.1	50.9	276 28.7	16.5	Atria	107 07.3	S 69 04.7		
20	303 40.0		91 04.0	58.3	183 19.0	26.5	162 35.1	50.8	291 31.3	16.4	Avior	234 14.5	S 59 35.4		
21	318 42.5		106 05.8	58.8	198 19.9	26.2	177 37.1	50.6	306 33.9	16.3	Bellatrix	278 21.3	N 6 22.5		
22	333 44.9		121 07.6	59.3	213 20.8	26.0	192 39.1	50.5	321 36.6	16.3	Betelgeuse	270 50.5	N 7 24.8		
23	348 47.4		136 09.5	59.9	228 21.7	25.7	207 41.1	50.4	336 39.2	16.2	Canopus	263 51.8	S 52 42.2		
Mer. Pass. 23 44.7		v 1.8 d 0.5		v 0.9 d 0.3		v 2.0 d 0.1		v 2.6 d 0.1		Capella	280 19.7	N46 01.4			
SHA		Mer. Pass.								Deneb	49 24.6	N45 22.8			
Venus	147 29.3	13 56								Denebola	182 23.8	N14 25.4			
Mars	239 51.6	07 47								Diphda	348 45.6	S 17 50.2			
Jupiter	218 58.8	09 11													
Saturn	347 49.9	00 37								Dubhe	193 39.9	N61 36.3			
Using the pages. GHA of a star = GHA Aries + SHA. Between the hours add the increment for the minutes and seconds (15° an hour for the Sun and planets, 15° 02.5' for Aries, 14° 19.0' for the Moon), then v × the fraction of the hour; change the declination by d × the fraction, in the direction the column is going.															
Mer. Pass. is the UT at which the body crosses the Greenwich meridian; for another longitude add 4 minutes for every degree west (subtract for east).															
Elnath														278 00.0	N28 37.8
Eltanin														90 41.5	N51 29.4
Enif														33 37.2	N10 00.1
Fomalhaut														15 12.6	S 29 28.7
Gacrux														171 50.7	S 57 15.8
Gienah														175 42.5	S 17 41.4
Hadar														148 34.6	S 60 30.2
Hamal														327 49.3	N23 35.4
Kaus Australis														83 30.6	S 34 22.4
Kochab														137 20.7	N74 02.9
Markab														13 28.2	N15 21.2
Menkar														314 04.4	N 4 11.8
Menkent														147 56.3	S 36 30.1
Miaplacidus														221 38.7	S 69 49.4
Mirfak														308 25.9	N49 57.3
Nunki														75 45.9	S 26 15.8
Peacock														53 03.2	S 56 39.0
Pollux														243 15.7	N27 57.6
Procyon														244 49.5	N 5 09.5
Rasalhague														95 57.3	N12 32.6
Regulus														207 33.2	N11 50.2
Rigel														281 02.4	S 8 10.0
Rigel Kentaurus														139 38.9	S 60 56.9
Sabik														102 01.2	S 15 45.4
Schedar														349 28.9	N56 41.1
Shaula														96 08.5	S 37 07.5
Sirius														258 25.0	S 16 44.9
Spica														158 21.1	S 11 18.0
Suhail														222 45.6	S 43 32.2
Vega														80 32.2	N38 48.8
Zubenelgenubi														136 54.7	S 16 09.1
Polaris														312 58.1	N89 22.4

UT	SUN			MOON					
	GHA	Dec		GHA	v	Dec	d	HP	
00	181 57.1	S 0 23.3		32 25.8	14.1	S 12 13.1	13.7	56.0	
01	196 57.3	24.2		46 58.9	14.1	S 11 59.4	13.8	56.1	
02	211 57.5	25.2		61 32.0	14.1	45.6	13.8	56.1	
03	226 57.7	26.2		76 05.1	14.1	31.8	13.9	56.1	
04	241 57.9	27.2		90 38.2	14.1	17.9	14.0	56.2	
05	256 58.1	28.1		105 11.3	14.2	03.9	14.0	56.2	
06	271 58.4	S 0 29.1		119 44.5	14.2	S 10 49.9	14.1	56.2	
07	286 58.6	30.1		134 17.7	14.2	35.8	14.1	56.2	
08	301 58.8	31.0		148 50.9	14.2	21.7	14.2	56.3	
09	316 59.0	32.0		163 24.1	14.2	07.5	14.3	56.3	
10	331 59.2	33.0		177 57.3	14.2	S 9 53.2	14.3	56.3	
11	346 59.5	34.0		192 30.5	14.2	38.9	14.4	56.3	
12	1 59.7	S 0 34.9		207 03.8	14.3	S 9 24.5	14.4	56.4	
13	16 59.9	35.9		221 37.0	14.3	10.1	14.5	56.4	
14	32 00.1	36.9		236 10.3	14.3	S 8 55.6	14.5	56.4	
15	47 00.3	37.9		250 43.6	14.3	41.1	14.6	56.5	
16	62 00.5	38.8		265 16.8	14.3	26.5	14.6	56.5	
17	77 00.8	39.8		279 50.1	14.3	11.9	14.7	56.5	
18	92 01.0	S 0 40.8		294 23.4	14.3	S 7 57.2	14.7	56.5	
19	107 01.2	41.8		308 56.7	14.3	42.4	14.8	56.6	
20	122 01.4	42.7		323 30.0	14.3	27.7	14.8	56.6	
21	137 01.6	43.7		338 03.3	14.3	12.8	14.9	56.6	
22	152 01.9	44.7		352 36.6	14.3	S 6 58.0	14.9	56.6	
23	167 02.1	45.6		7 09.9	14.3	43.1	15.0	56.7	
SD 15.9 d 1.0				SD 15.4					

Day	SUN			MOON				
	Eqn. of Time	Mer. Pass.		Mer. Pass.	Age	Phase		
	00h	12h	h m	Upper	Lower	d	%	
24	07 48	07 59	11 52	22 30	10 08	13	95	

- UT is UTC with DUT1 = 0 (CONVENTIONS 6); the printed almanac's argument is UT1. The difference is up to 0.23' of GHA and 1 s of any time.
- v: the excess of the hourly increase of GHA over 15° (planets, mean over the day) or 14° 19.0' (Moon, from each hour to the next). d: the hourly change of declination, printed without sign; take the sign from the trend of the column.
- Stars' SHA and Dec, the planets' SHA and magnitudes, SD, the Moon's age and percentage illuminated are for 12h UT of the date. The stars stay within 0.02' of their row all day; Polaris, which the printed almanac tabulates separately, moves up to 0.8' a day in SHA and is within 0.4' of its row.
- Equation of time: apparent minus mean solar time at Greenwich, in minutes and seconds; negative (shaded on the page, a minus sign in plain text) when the Sun crosses the meridian after 12h.
- Twilight, sunrise, sunset, moonrise and moonset: LMT at the Greenwich meridian (= UT) for a sea-level observer. Sun's centre at -50' (sunrise, sunset), -6° (civil) and -12° (nautical); the Moon's upper limb on the horizon with 34' of refraction (CONVENTIONS 13.3).
- □ above the horizon all day; ■ below it (or below the twilight altitude) all day; /// twilight all night; 24 hh mm: on the following date; -00 mm: minutes before 00h; --: not on the date nor the next; n/a: outside the ephemeris coverage.
- Simulation and analysis workbench. Not a navigation instrument.

Lat.	Twilight		Sunrise	Moonrise	
	Naut.	Civil		24	25
N 72	03 10	04 39	05 47	18 06	17 31
N 70	03 29	04 46	05 48	17 54	17 28
N 68	03 43	04 52	05 48	17 45	17 26
N 66	03 54	04 57	05 48	17 37	17 24
N 64	04 03	05 01	05 48	17 30	17 22
N 62	04 11	05 04	05 49	17 25	17 21
N 60	04 17	05 07	05 49	17 19	17 19
N 58	04 23	05 10	05 49	17 15	17 18
N 56	04 28	05 12	05 49	17 11	17 17
N 54	04 32	05 14	05 49	17 07	17 16
N 52	04 36	05 15	05 49	17 04	17 15
N 50	04 39	05 17	05 49	17 01	17 15
N 45	04 46	05 20	05 49	16 54	17 13
N 40	04 51	05 22	05 49	16 49	17 11
N 35	04 55	05 24	05 49	16 44	17 10
N 30	04 58	05 26	05 49	16 40	17 09
N 20	05 02	05 27	05 49	16 33	17 07
N 10	05 04	05 28	05 49	16 26	17 06
0	05 04	05 28	05 49	16 20	17 04
S 10	05 03	05 27	05 48	16 14	17 03
S 20	05 00	05 26	05 48	16 07	17 01
S 30	04 55	05 23	05 47	16 00	16 59
S 35	04 52	05 21	05 47	15 55	16 58
S 40	04 48	05 19	05 46	15 50	16 57
S 45	04 42	05 16	05 45	15 45	16 55
S 50	04 34	05 12	05 45	15 38	16 54
S 52	04 31	05 11	05 44	15 35	16 53
S 54	04 27	05 09	05 44	15 31	16 52
S 56	04 22	05 06	05 43	15 27	16 51
S 58	04 17	05 04	05 43	15 23	16 50
S 60	04 10	05 01	05 42	15 18	16 49

Lat.	Sunset	Twilight		Moonset	
		Civil	Naut.	24	25
N 72	17 54	19 02	20 29	01 08	03 27
N 70	17 54	18 55	20 12	01 30	03 36
N 68	17 54	18 50	19 58	01 47	03 43
N 66	17 54	18 45	19 48	02 01	03 49
N 64	17 54	18 41	19 39	02 12	03 53
N 62	17 54	18 38	19 31	02 22	03 58
N 60	17 54	18 35	19 25	02 30	04 01
N 58	17 54	18 33	19 19	02 37	04 04
N 56	17 54	18 31	19 15	02 43	04 07
N 54	17 54	18 29	19 11	02 49	04 10
N 52	17 54	18 27	19 07	02 54	04 12
N 50	17 54	18 26	19 04	02 59	04 14
N 45	17 54	18 23	18 57	03 08	04 18
N 40	17 54	18 21	18 52	03 16	04 22
N 35	17 54	18 19	18 49	03 23	04 25
N 30	17 54	18 18	18 46	03 29	04 28
N 20	17 54	18 16	18 42	03 40	04 33
N 10	17 55	18 16	18 40	03 49	04 37
0	17 55	18 16	18 40	03 57	04 41
S 10	17 56	18 17	18 41	04 05	04 45
S 20	17 56	18 18	18 44	04 14	04 49
S 30	17 57	18 21	18 49	04 24	04 53
S 35	17 58	18 23	18 53	04 30	04 56
S 40	17 59	18 26	18 57	04 36	04 59
S 45	17 59	18 29	19 03	04 43	05 02
S 50	18 00	18 33	19 11	04 52	05 06
S 52	18 01	18 35	19 15	04 56	05 08
S 54	18 01	18 37	19 19	05 01	05 10
S 56	18 02	18 39	19 24	05 06	05 12
S 58	18 03	18 42	19 29	05 11	05 15
S 60	18 03	18 45	19 36	05 17	05 17